

Logic

II

15

The Kets of  
Naucratis.

\* Egyptian Exploration  
Fund. Third  
Memoir. p.

mention an incontestable instance of it; - incontestable, at least,  
by any fair-minded competent to deal with the problem. Prof.  
Flinders Petrie, whose reasoning powers I had admired long  
before his <sup>other</sup> great scientific qualities had been proved, among which  
his great exactitude and circumspection as a metrologist con-  
cerns us here, <sup>exhumed</sup> at the ancient trading town of Naucratis,  
no less than 158 <sup>balance =</sup> weights having the Egyptian ket as their unit.\*  
The great majority of them are of basalt and syenite, material so  
unchangeable that the corrections needed to bring them to their original  
values are small. I shall deal only with 144 of them from <sup>each of</sup> which  
Mr. Petrie has calculated the value of the ket to a tenth of a Troy  
grain. Since these values range all the way from 137 to 152  
grains, it is evident that <sup>the weights</sup> ~~they~~ <sup>were intended to be copies of</sup> ~~represent~~ several different  
standards, probably four or five; for there <sup>would be</sup> is no use of  
<sup>a balance,</sup> ~~weighing~~ <sup>could</sup> if one can detect the errors of the <sup>balance =</sup> weights by  
simply half "hefting" them, and comparing them with one's  
memory of the standard weight. ~~Therefore~~ Considering that these



Logic  
II  
16

~~classes~~ weights are small, and were therefore used for weighing costly or even precious matter; our knowledge of <sup>The practice</sup> ~~an amount~~ of weighing among the ancients <sup>gives us ground for thinking it likely that about half</sup> ~~enables us to guess with some confidence~~ the weights would depart from their virtual standards by more, and about half by less, than, say,  $\frac{1}{4}$  or  $\frac{1}{5}$  tenths of one per cent, which, upon a bet, would be <sup>from half to two thirds of</sup> ~~a little more than half~~ a grain. Now the whole interval here is  <sup>$\frac{144}{1000}$</sup>   ~~$\frac{144}{1000}$~~  grains, and between 136.8 grains to 151.3 grains, there is no case of an interval of more than a third of a grain not represented by any weight <sup>among</sup> of the 144. To a person thoroughly familiar with the theory of errors this shows that there must be four or five different standards to which different ones aim to conform. In order that every reader may see that this is the case, I give a diagram upon which the abscissas are grains <sup>whole</sup> and the ordinates <sup>represent</sup> ~~are~~ the numbers of weights among the 144 which are either exactly of the weight represented by the ~~the~~ abscissas of the points plotted or exceed them

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Logic  
II  
17

weight by less than half a grain. The points plotted in red with circles about them are these actual observations. In order to ~~have~~ represent these observations, I have adopted the following rough-and-ready theory; for to make elaborate calculations would, from every point of view, be a waste of time. I have assumed that there were five different standards; and that the ~~weights~~ weights depart from their standards according to the probability curve (which will be explained in a subsequent chapter); and that the probable error of a single weight is  $\frac{5}{8}$  grain. I assume that of the 144 weights

|    |      |                                      |               |
|----|------|--------------------------------------|---------------|
| 36 | were | designed to conform to a standard of | 139.2 grains  |
| 25 | "    | "                                    | 139.0 grains  |
| 26 | "    | "                                    | 142.9 grains  |
| 23 | "    | "                                    | 144.7 grains  |
| 34 | "    | "                                    | 146.95 grains |
|    |      |                                      | 149.7 grains  |

besides,  
I have drawn on the diagram curves in blue, which cut the ordinates at the red dots in points whose ordinates show



Logic

II

18

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how many weights of each class there would be, upon this theory, in every half grain. There is also a brown curve ~~which~~ whose ordinates are the sums of those of the blue ones and which would ~~not~~ pass through every red dot if it were to represent the observations with mechanical accuracy. I repeat that this theory has not been the subject of any but the simplest calculations. It is obvious that some such theory must be true, but to decide how near my theory probably comes to the true theory or how it ought to be ~~able~~ <sup>modified</sup> ~~is~~ would be a very intricate problem for the solution of which the data are probably insufficient. It does not concern us here; our objects being merely to make it clear that truly natural classes may, and undoubtedly often do merge into one another inextricably.

It is, I think, pretty certain that there were as many as five standards. Before the adoption of the metric system, every ~~part~~ city throughout the greater part, if not all, the continent of Europe



Logic  
II  
19

had its own pounds, like its own patrics. See the article "pound" in the Century Dictionary, which ~~was~~ <sup>whose values were</sup> based on a list of some three hundred of such pounds known to me, ~~with their~~ <sup>is</sup> a list now kept in MS. in the Astor Library. That the same state of things must have been true in ancient Egypt may be inferred from the looseness of the tie which bound the <sup>different</sup> provinces of that empire together. Even their religions were different; so that a fortiori their kets would be so. Besides, none of the kets carry any authoritative mark; which is pretty conclusive proof that the central government did not intervene. It is, therefore, probable that the five standards were those of five towns with which Naucratis carried on trade. Yet virtual standards may be created in other ways. For example <sup>where</sup> ~~when~~ government does not insure uniformity in weights, it is usual for buyers to bring their own weights. <sup>It</sup> ~~There~~ would thus naturally happen that some <sup>balance-</sup>weights would be manufactured for the use of buyers, and others for the use of sellers; and thus there



Logic  
II  
20

would naturally be a tendency to the crystallization of a heavier and a lighter norm.

As for my assumption that the departures of the single weights from their virtual standards conform to the probability curve, it was only adopted as a ready way of imparting definiteness to the problem. Rich as is the store of data given by Petrie, it is ~~insufficient~~ insufficient, apparently, for determining the true law of these departures. If the workmen were sufficiently skillful. (as I believe they would be) ~~these departures~~ the departures would follow the probability curve. But if they were unskillful, it would be desirable to ascertain <sup>by</sup> what process the weights were made. The weights, being of stone, are not loaded; so that the adjustment was made by grinding, exclusively? Did the workman <sup>then</sup> have a balance by his side, or did he finish the weight by guess-work? In the latter case, inspection (and some sort of inspection there must, in this case, have been)



Logie  
II  
21

would reject all weights outside a certain "tolerance," as it is called in coinage. Those that were too light would have to be thrown away. They would lie in a heap, until they <sup>however,</sup> appeared to deceive a future archeologist. (Petrie's weights, <sup>not lighter,</sup> are somewhat heavier <sup>than</sup> independent evidence would lead us to believe the set to have been.) Those that were too heavy would be reground, but would for the most part still be rather heavier than the standard. The consequence would be that the curve would be cut ~~sh~~ down vertically at two ordinates (equally distant, perhaps, from the standard,) while the ordinate of its maximum would be at the right of that of the standard. If the workman had a balance at hand, and frequently used it during the process of adjustment, the form of the error-curve would depend upon the construction of the balance. If it were like a modern balance, <sup>so as to show,</sup> ~~which shows,~~ not only that one mass is greater than another, but also whether it is much or little greater, the workman would ~~be~~ keep in one pan



Logie  
II  
22

a weight ~~equal~~ to the maximum value that he proposed himself as permissible for the weight he was making; and in all his successive grindings would be aiming at that. The consequence would be a curve  concave upwards and stopping abruptly at its maximum ordinate; — a form easily manageable by a slight modification of the method of least squares. [I have drawn the curve in the upper left hand corner of the plate.] But most of the balances shown upon the Egyptian monuments are provided with stops or other contrivances which would be needless if the balances were not top-heavy.\* Now a top-heavy balance will not show that two weights are equal, otherwise than by remaining with either ~~end~~ end down which may be down. It only shows when, ~~or when~~ a weight is ~~not~~ ~~indeed~~ being already in one pan, a decidedly heavier weight is placed in the other. The workman using such a balance would have no warning that he was approaching the limit, and would be unable to aim at

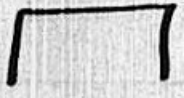
such balances, working automatically, are in use in all the mints of the civilized world, for throwing out light and heavy coins.



# Logic

## II

23

any definite value, but would have to grind away blindly, trying his weight every time he had ground off about as much as the whole range of variation which he proposed to allow himself. If he always ground off precisely the same amount between successive tryings of his weight, he would be just as likely to grind below his maximum by any one fraction of the amount taken off at a grinding as by any other; so that his error curve would be a horizontal line cut off by vertical ordinates; thus, . But since there would be a variability in the amount taken off between the trials, the ~~left hand~~ <sup>corner</sup> ~~side~~ of the curve would show a sort of moulding

with a contrary flexure, thus: . [I have drawn it in the upper right hand corner of the plate.]

I hope this long digression, (which will be referred to with some interest when we come to study the theory of errors) will not have caused the reader to forget that we were engaged in tracing out some of the consequences of understanding the term "natural" or "real" class to mean a class the existence of whose members is due to a common and peculiar final cause. This, as I was saying, is a wide-spread error to think that a "final cause" is necessarily a purpose, & p. is

It must be admitted that the distribution of Petrie's kets is suggestive of this sort of curve, or rather of a modification of it due to a middling degree of skill.

Final Causation



merely that form of final cause which is most familiar to our experience. The signification of the phrase "final cause" must be determined by its use in the statement of Aristotle that all causation divides into two grand branches, the efficient, or forceful; and the ideal, or final. If we are to conserve the truth of that statement, we must understand by final causation that mode of bringing facts about according to which a general description of result is ~~made~~ made to <sup>about</sup> come, quite irrespective of any compulsion for it to <sup>come about</sup> appear in this or that particular way; although the means may be adapted to the end. The general result may be brought about at one time in one way, and at another time in another way. Final causation does not determine in what particular way it is to be brought about, but only that the general result shall have a certain general character. Efficient causation, on the other hand, is a compulsion determined by the particular